

S-DES CBC MODE

Key : 0111111101
IV : 10101010
Plaintext : 00000001 00100011

Encryption

Ciphertext: 11110100 00001011

Decryption

Plaintext : 00000001 00100011

S-DES COUNTER (CTR) MODE

Key : 0111111101

Counter : 00000000

Plaintext : 00000001 00000010 00000100

Encryption

Ciphertext: 00111000 01001111 00110010

Decryption

Plaintext : 00000001 00000010 00000100

----- RESTART: C:/Users/GOD/AppData/LOC

Step 1: Find p and q

$p = 59$

$q = 61$

Step 2: Calculate $\phi(n)$

$\phi(n) = 3480$

Step 3: Find private exponent d

$\gcd = 1$

$d = 3031$

Verification:

$(e * d) \bmod \phi(n) = 1$

Public Key = (31 , 3599)

Private Key = (3031 , 3599)

Original RSA Keys

$n = 3233$

Old Public Key = (17, 3233)

Old Private Key = (2753, 3233)

Private key leaked!

Leaked $d = 2753$

Bob generates a new key

New Public Key = (7, 3233)

New Private Key = (1783, 3233)

Is it safe?

NO

Reason:

The modulus n is still the same.

If the attacker can factor n ,

the attacker can calculate the new private key.

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----- RESTART. C:/Users/GOD/AppData/LO

RSA Public Key

$e = 31$

$n = 3599$

Known plaintext block:

$m = 118$

Step 1: Calculate GCD

$\gcd(m, n) = 59$

Step 2: Factor n

$p = 59$

$q = 61$

Step 3: Calculate $\phi(n)$

$\phi(n) = 3480$

Step 4: Find private key

$d = 3031$

Private Key = (3031 , 3599)

Public values:

$a = 5$

$q = 23$

Secret values:

Alice's secret $x = 6$

Bob's secret $y = 15$

Values exchanged:

Alice sends: 2

Bob sends : 7

Shared key:

Alice's key = 4

Bob's key = 16

Key agreement failed.

SHA-3 Lane Analysis

State size : 1600 bits
Block size : 1024 bits
Capacity : 576 bits
Total lanes : 25
Capacity lanes : 9

After first message block:

Nonzero lanes = 16

Zero lanes = 9

After second message block:

Nonzero lanes = 16

Zero lanes = 9

Capacity lanes still zero: 9

Some capacity lanes are still zero.

CBC-MAC Demonstration

X = 00111100

MAC T = 01100110

X XOR T = 01011010

Original message:

X = 00111100

MAC(X) = 01100110

Forged message:

X || (X XOR T) = 00111100 01011010

MAC(forged) = 01100110

Attack successful!

Both messages have the same CBC-MAC.

===== RESIARI: C:/Users/God/AppData/LOC

RSA Codebook

A -> 0
B -> 1
C -> 1752
D -> 1211
E -> 1387
F -> 3086
G -> 824
H -> 2369
I -> 2041
J -> 1972
K -> 1096
L -> 3061
M -> 1730
N -> 47
O -> 2549
P -> 3031
Q -> 134
R -> 908
S -> 2100
T -> 615
U -> 3023
V -> 1188
W -> 2558
X -> 2037
Y -> 1639
Z -> 2211

Ciphertext:

[2369, 1387, 3061, 3061, 2549]

Recovered plaintext:

HELLO

Enter plaintext: hello

Padded Data:

[104, 101, 108, 108, 111, 1, 0, 0]

ECB Mode:

[98, 111, 102, 102, 101, 11, 10, 10]

CBC Mode:

[99, 109, 101, 98, 6, 102, 111, 104]

CFB Mode:

[99, 109, 101, 98, 6, 102, 111, 104]